

Lab 18 Worksheet — Printer Installation, Configuration and the Laser Imaging Process

Name: ____ Date: ____

Exam objective: Install and share a printer, configure its settings and security, and map each stage of the laser imaging process to the print defect its failure produces (Core 1 objectives 3.6 and 3.7).

Record as you go

Step	What you did	What you observed
1	Unbox and site the printer correctly: adequate clearance for paper paths and covers, a suitable power source and the right connection — USB, Ethernet or wireless.	
2	Install the printer driver, choosing between PCL, developed by HP and most common, and PostScript, developed by Adobe and used in graphics and industrial work.	
3	Add the printer in Windows and print a test page to confirm the driver and connection work end to end.	
4	Configure the default settings users ask about most: duplex printing, orientation, paper tray selection and print quality.	
5	Share the printer and record the difference between a printer share, where a workstation shares its own printer and must stay on, and a print server, which is a dedicated always-on host.	
6	Configure printer security: user authentication so only authorised staff may print, badging where supported, audit logging and secured or held print release.	
7	Record the seven stages of the laser imaging	

Step	What you did	What you observed
	process in order: processing, charging, exposing, developing, transferring, fusing and cleaning.	
8	Map stage failures to defects: a scratched imaging drum produces vertical lines down every page, and a failed primary corona produces entirely blank or entirely black pages.	
9	Continue the map: a failed fuser leaves toner that smudges when rubbed because it was never melted onto the paper, and failed cleaning produces repeated ghost images at the drum's circumference.	
10	Record the maintenance kit contents — fuser unit, transfer roller, paper feed and separation rollers and pickup rollers — and note that the printer must be calibrated after fitting one.	
11	Compare the other printer types on cost, speed and use: inkjet for affordable colour, impact for multi-part carbon forms, thermal for receipts, and 3D printers using FDM filament or SLA resin.	
12	Record the safety rules: avoid inhaling toner, never use a normal vacuum on toner because it passes through the filter, and let the fuser cool before touching it as it operates above 180 degrees Celsius.	

Verification

Success criterion: The printer prints a test page and is shared successfully, your defect map correctly links at least four print defects to the specific laser stage that failed, and the safety rules include the toner vacuum and hot fuser warnings.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
